Elastic Collisions

Answer key

Conventions used throughout

• **Symbols (from the lessons):** $\vec{p} = m\vec{v}$ is momentum (kg·m/s). A prime (′) means *after* the collision; no prime means before. Subscripts name the object.

• **Method (from the lesson):** draw a before and after diagram, choose the + direction, then write $\vec{p}_{tot} = \vec{p}_{tot}'$, that is $m_1v_1 + m_2v_2 = m_1v_1' + m_2v_2'$.

• **Signs:** in one dimension every velocity carries its sign in the working ($v = -3.0$ m/s). The direction is written in square brackets only in the final answer.

• **Elastic collisions (from the lesson):** kinetic energy is conserved as well as momentum: $\frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 = \frac{1}{2}m_1v_1'^2 + \frac{1}{2}m_2v_2'^2$.

• **Head-on elastic collisions:** the lesson's two equations are used once they have been given: $v_1' = ((m_1 - m_2)/(m_1 + m_2))v_1 + (2m_2/(m_1 + m_2))v_2$ and $v_2' = ((m_2 - m_1)/(m_1 + m_2))v_2 + (2m_1/(m_1 + m_2))v_1$.

• **Rounding:** the practice questions give data to 1 figure (3 kg, 4 m/s). The key keeps 2 figures in those answers, as the worksheet's answers do.

• **Solve in letters first:** each unknown is isolated as a formula before any numbers go in.

• $g = 9.81$ m/s^2. Every step keeps full precision; only final answers are rounded.

EXAMPLE 1

Fast and slow

Two carts of equal mass collide as shown below. Predict what their final velocities will be, then observe and record the results

Let's calculate this outcome.

Conservation of Momentum: Sub in the known values and solve for final velocity:

Conservation of Kinetic Energy: Sub in your known values, simplify, and then combine the momentum equation to solve for $\vec{v}_A'$ and $\vec{v}_B'$. (You can drop your units.

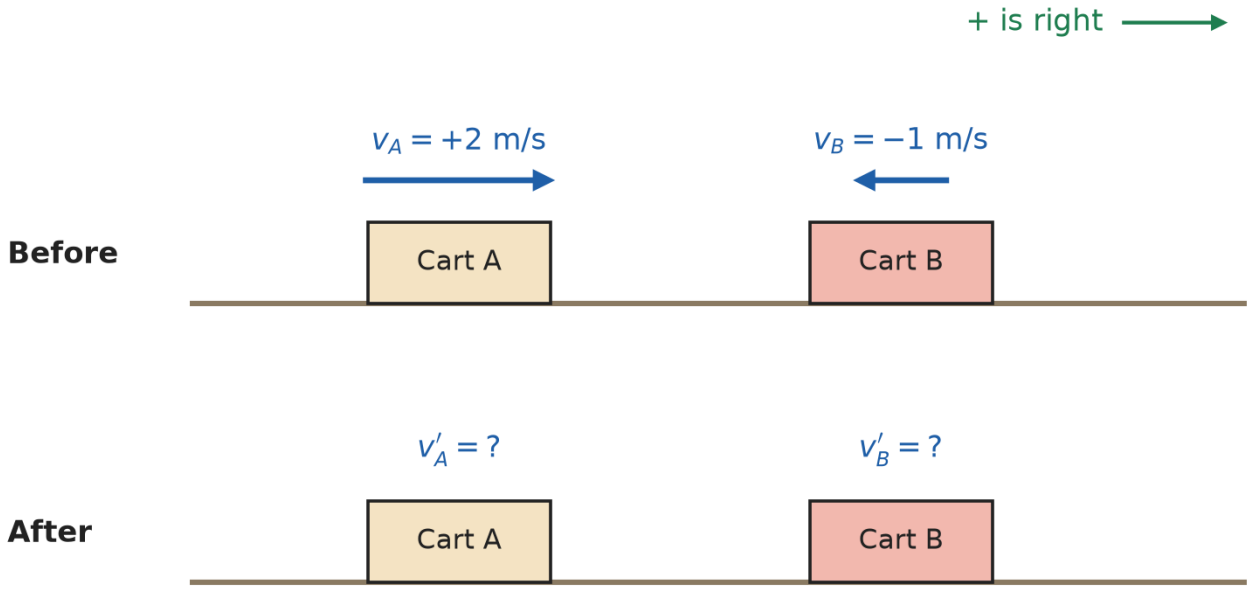
GIVEN

$m_A = m_B = m$ $\quad v_A = +2$ m/s $\quad v_B = -1$ m/s

$v_A' = ?$ $\quad v_B' = ?$

Sign convention: right is +. Cart B moves left, so $v_B = -1$ m/s.

Before and after (+ is right)



Step 1 – Conservation of momentum

The masses are equal, so m divides out of every term.

Formula: $m_A v_A + m_B v_B = m_A v'_A + m_B v'_B$

Put in the values: $m(2) + m(-1) = m v'_A + m v'_B$

Divide every term by m: $1 = v'_A + v'_B \qquad (1)$

This is one equation with two unknowns, so it cannot be solved by itself.

Step 2 – Conservation of kinetic energy

Every term has ½m, so it divides out too.

Formula: $\frac{1}{2}m_A v_A^2 + \frac{1}{2}m_B v_B^2 = \frac{1}{2}m_A v'^2_A + \frac{1}{2}m_B v'^2_B$

Put in the values: $\frac{1}{2}m(2)^2 + \frac{1}{2}m(-1)^2 = \frac{1}{2}m v'^2_A + \frac{1}{2}m v'^2_B$

Divide every term by $\frac{1}{2}m$: $5 = v'^2_A + v'^2_B \qquad (2)$

Step 3 – Combine the two equations

Isolate v_B' in (1) and put it into (2).

From (1): $v_B' = 1 - v_A'$

Put into (2): $5 = v_A'^2 + (1 - v_A')^2$

Expand: $5 = v_A'^2 + 1 - 2v_A' + v_A'^2$

Collect and divide by 2: $v_A'^2 - v_A' - 2 = 0$

Factor: $(v_A' - 2)(v_A' + 1) = 0 \quad \Rightarrow \quad v_A' = 2 \text{ or } v_A' = -1$

v_A' = +2 m/s is cart A's velocity before the collision (the answer for "no collision"). The carts do collide, so v_A' = −1 m/s.

Step 4 – Cart B

From (1): $v_B' = 1 - v_A' = 1 - (-1) = +2$ m/s

The two equal carts have traded velocities.

FINAL ANSWER

v_A' = −1 m/s; $\vec{v}_A'$ = 1 m/s [left]

v_B' = +2 m/s; $\vec{v}_B'$ = 2 m/s [right]

SPECIAL CASES

The elastic collision equations in three special cases

For each of the cases below, how would the above equations be different? Make the appropriate substitutions and work them out:

A) One of the objects is initially at rest

B) The objects have the same mass

C) One object is much, much heavier (assume that in comparison to m_1, $m_2 \approx 0$)

The two equations for head-on elastic collisions, from the lesson:

$$v_1' = \left(\frac{m_1 - m_2}{m_1 + m_2}\right)v_1 + \left(\frac{2m_2}{m_1 + m_2}\right)v_2$$

$$v_2' = \left(\frac{m_2 - m_1}{m_1 + m_2}\right)v_2 + \left(\frac{2m_1}{m_1 + m_2}\right)v_1$$

A) One of the objects is initially at rest

Let object 2 be the one at rest: $v_2 = 0$. Every v_2 term disappears.

$$\text{Put in } v_2 = 0: \quad v_1' = \left(\frac{m_1 - m_2}{m_1 + m_2}\right)v_1$$

$$v_2' = \left(\frac{2m_1}{m_1 + m_2}\right)v_1$$

B) The objects have the same mass

Put in $m_1 = m_2 = m$. Then $m_1 - m_2 = 0$ and $2m/(m + m) = 1$.

$$\text{Object 1:} \quad v_1' = \left(\frac{0}{2m}\right)v_1 + \left(\frac{2m}{2m}\right)v_2 = v_2$$

$$\text{Object 2:} \quad v_2' = \left(\frac{0}{2m}\right)v_2 + \left(\frac{2m}{2m}\right)v_1 = v_1$$

The objects trade velocities, as the two carts did in Example 1.

C) One object is much, much heavier (assume that in comparison to m_1, $m_2 \approx 0$)

Put in $m_2 = 0$.

Object 1 (heavy): $$v_1' = \left(\frac{m_1 - 0}{m_1 + 0}\right)v_1 + \left(\frac{0}{m_1}\right)v_2 = v_1$$

Object 2 (light): $$v_2' = \left(\frac{0 - m_1}{m_1 + 0}\right)v_2 + \left(\frac{2m_1}{m_1 + 0}\right)v_1 = 2v_1 - v_2$$

The heavy object carries on unchanged. The light object bounces off it: if the light object starts at rest it leaves at twice the heavy object's velocity.

FINAL ANSWER

A) $v_1' = ((m_1 - m_2)/(m_1 + m_2))v_1$, $v_2' = (2m_1/(m_1 + m_2))v_1$

B) $v_1' = v_2$ and $v_2' = v_1$ (they trade velocities)

C) $v_1' = v_1$ and $v_2' = 2v_1 - v_2$

DROPPING BALLS

How many times heavier should the large ball be?

Watch the demonstration. The large ball hits the ground first and bounces back up, hitting the smaller ball. After this collision, the large ball stops dead and the small ball bounces really high.

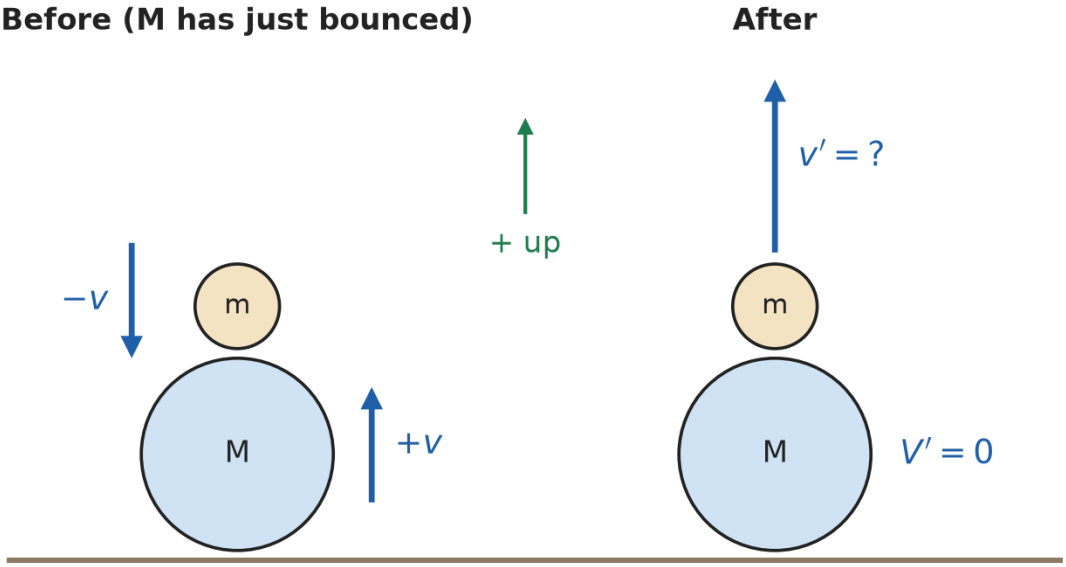
Goal: How many times heavier should the large ball be?

Momentum: Using conservation of momentum, solve for v'/v

Kinetic Energy: Using conservation of kinetic energy, solve for v'/v

Now solve for M in terms of m:

The two balls just before and just after they collide (+ is up)



From the worksheet's picture: before, M moves up at +v and m moves down at −v. After, M is at rest (V′ = 0) and m moves up at v′.

Momentum Using conservation of momentum, solve for v’/v

Up is +. After the collision the large ball has no momentum.

Formula: $Mv_M + mv_m = Mv'_M + mv'_m$

Put in the velocities: $M(+v) + m(-v) = M(0) + mv'$

Factor the left side: $(M - m)\,v = mv'$

Divide both sides by mv: $\dfrac{v'}{v} = \dfrac{M - m}{m}$

Kinetic Energy Using conservation of kinetic energy, solve for v’/v

The speeds are squared, so the signs do not matter here.

Formula: $\dfrac{1}{2}Mv^2 + \dfrac{1}{2}mv^2 = \dfrac{1}{2}M(0)^2 + \dfrac{1}{2}mv'^2$

Multiply by 2 and factor: $(M + m)\,v^2 = mv'^2$

Divide both sides by mv^2: $\dfrac{v'^2}{v^2} = \dfrac{M + m}{m}$

Square root both sides: $\dfrac{v'}{v} = \sqrt{\dfrac{M + m}{m}}$

Solve Now solve for M in terms of m:

Both expressions equal v′/v, so they equal each other. Square both sides to remove the root.

Set them equal: $\frac{M-m}{m} = \sqrt{\frac{M+m}{m}}$

Square both sides: $\frac{(M-m)^2}{m^2} = \frac{M+m}{m}$

Multiply both sides by m^2: $(M-m)^2 = m(M+m)$

Expand: $M^2 - 2Mm + m^2 = Mm + m^2$

Subtract $Mm + m^2$: $M^2 - 3Mm = 0$

Divide both sides by M: $M = 3m$

Check: with M = 3m, momentum gives v′/v = (3m − m)/m = 2 and kinetic energy gives √(4m/m) = 2. The small ball leaves at twice the speed, so it rises four times as high.

FINAL ANSWER

Momentum: v′/v = (M − m)/m

Kinetic energy: v′/v = √((M + m)/m)

M = 3m: the large ball should be three times as heavy as the small ball

EXAMPLE

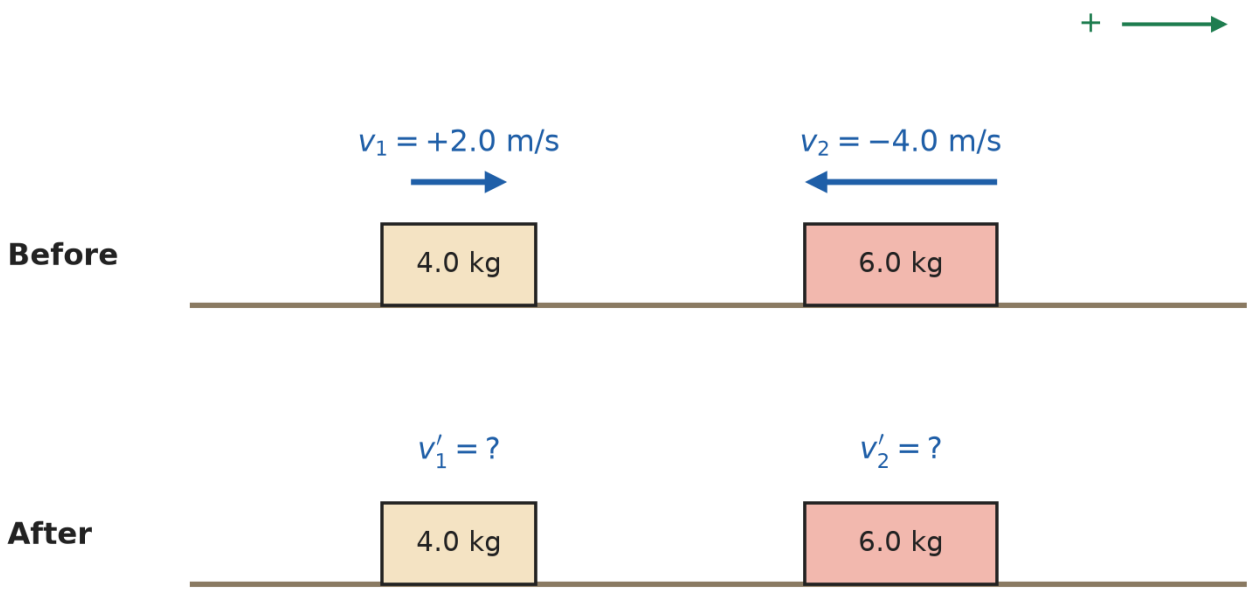
Two particles, elastic collision

A particle of mass 4.0 kg, initially moving with a velocity of 2.0 m/s collides elastically with a particle of mass 6.0 kg initially moving with a velocity of -4.0 m/s. What are the velocities of the two particles after the collision?

GIVEN

$m_1 = 4.0$ kg	$v_1 = +2.0$ m/s	$m_2 = 6.0$ kg
$v_2 = -4.0$ m/s	$v_1' = ?$	$v_2' = ?$

Before and after



Step 1 – The 4.0 kg particle

Use the lesson's equations for a head-on elastic collision.

Formula: $$v_1' = \left(\frac{m_1 - m_2}{m_1 + m_2}\right)v_1 + \left(\frac{2m_2}{m_1 + m_2}\right)v_2$$

Put in the numbers: $$v_1' = \left(\frac{4.0 - 6.0}{4.0 + 6.0}\right)(2.0) + \left(\frac{2(6.0)}{4.0 + 6.0}\right)(-4.0)$$

$$v_1' = (-0.2)(2.0) + (1.2)(-4.0) = -0.4 - 4.8 = -5.2 \text{ m/s}$$

Step 2 – The 6.0 kg particle

Formula: $v_2' = \left(\frac{m_2 - m_1}{m_1 + m_2}\right)v_2 + \left(\frac{2m_1}{m_1 + m_2}\right)v_1$

Put in the numbers: $v_2' = \left(\frac{6.0 - 4.0}{4.0 + 6.0}\right)(-4.0) + \left(\frac{2(4.0)}{4.0 + 6.0}\right)(2.0)$

$v_2' = (0.2)(-4.0) + (0.8)(2.0) = -0.8 + 1.6 = 0.8 \text{ m/s}$

FINAL ANSWER

v_1' = −5.2 m/s (the 4.0 kg particle now moves in the negative direction)

v_2' = +0.80 m/s (the 6.0 kg particle now moves in the positive direction)

Check: momentum before = (4.0)(2.0) + (6.0)(−4.0) = −16 kg·m/s; after = (4.0)(−5.2) + (6.0)(0.8) = −16 kg·m/s.

PRACTICE QUESTION 1

Two carts, total momentum zero

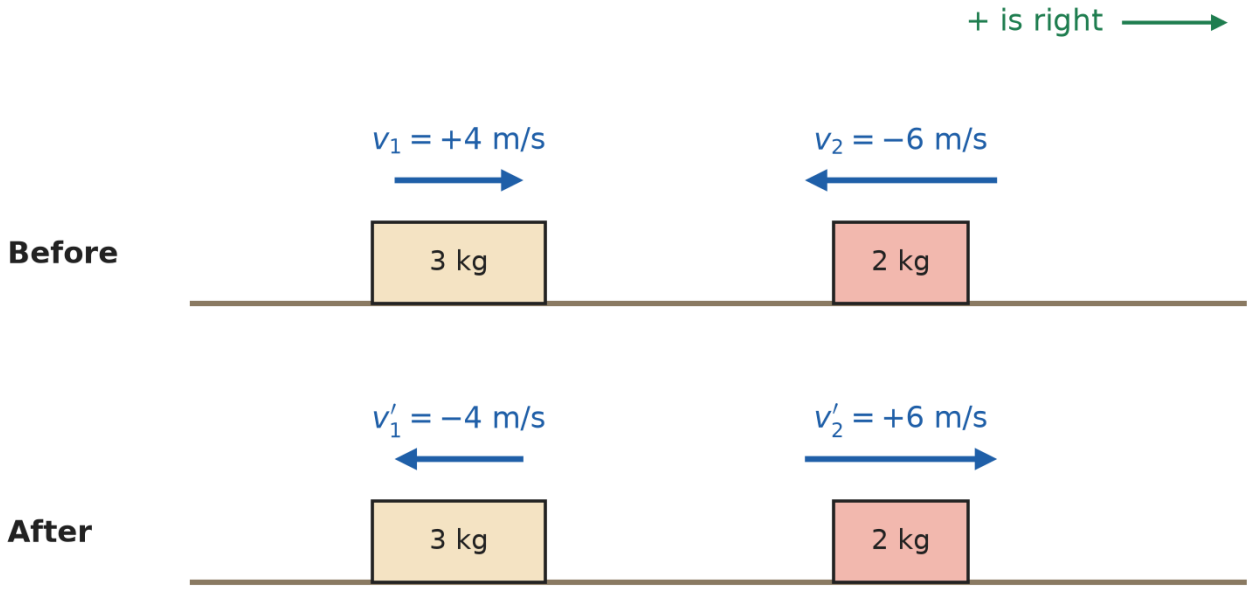
1. A 3 kg cart travelling to the right at 4 m/s undergoes an elastic collision with a 2 kg cart travelling to the left at 6 m/s. Determine the velocity of each cart after the collision. What did you notice about the initial total momentum?

GIVEN

$m_1 = 3$ kg	$v_1 = +4$ m/s	$m_2 = 2$ kg
$v_2 = -6$ m/s	$v_1' = ?$	$v_2' = ?$

Sign convention: right is +.

Initial and final state (+ is right)



Step 1 – The 3 kg cart on the left

Use the lesson's equations for a head-on elastic collision.

Formula: $$v_1' = \left(\frac{m_1 - m_2}{m_1 + m_2}\right)v_1 + \left(\frac{2m_2}{m_1 + m_2}\right)v_2$$

Put in the numbers: $$v_1' = \left(\frac{3-2}{3+2}\right)(4) + \left(\frac{2(2)}{3+2}\right)(-6)$$

$$v_1' = (0.2000)(4) + (0.8000)(-6) = 0.800 - 4.800 = -4.000 \text{ m/s}$$

Step 2 – The 2 kg cart on the right

Formula: $$v_2' = \left(\frac{m_2 - m_1}{m_1 + m_2}\right)v_2 + \left(\frac{2m_1}{m_1 + m_2}\right)v_1$$

Put in the numbers: $$v_2' = \left(\frac{2-3}{3+2}\right)(-6) + \left(\frac{2(3)}{3+2}\right)(4)$$

$$v_2' = (-0.2000)(-6) + (1.2000)(4) = 1.200 + 4.800 = 6.000 \text{ m/s}$$

The initial total momentum: (3)(+4) + (2)(−6) = 12 − 12 = 0. When the total momentum is zero in an elastic collision, each cart reverses direction and keeps its speed. That is the only way to keep the momentum at zero and the kinetic energy unchanged.

FINAL ANSWER

$v_1' = -4.0$ m/s; $\vec{v}_1' = 4.0$ m/s [left]

$v_2' = +6.0$ m/s; $\vec{v}_2' = 6.0$ m/s [right]

The initial total momentum is zero, so each cart leaves with its own speed, reversed

PRACTICE QUESTION 2

Two objects of equal mass

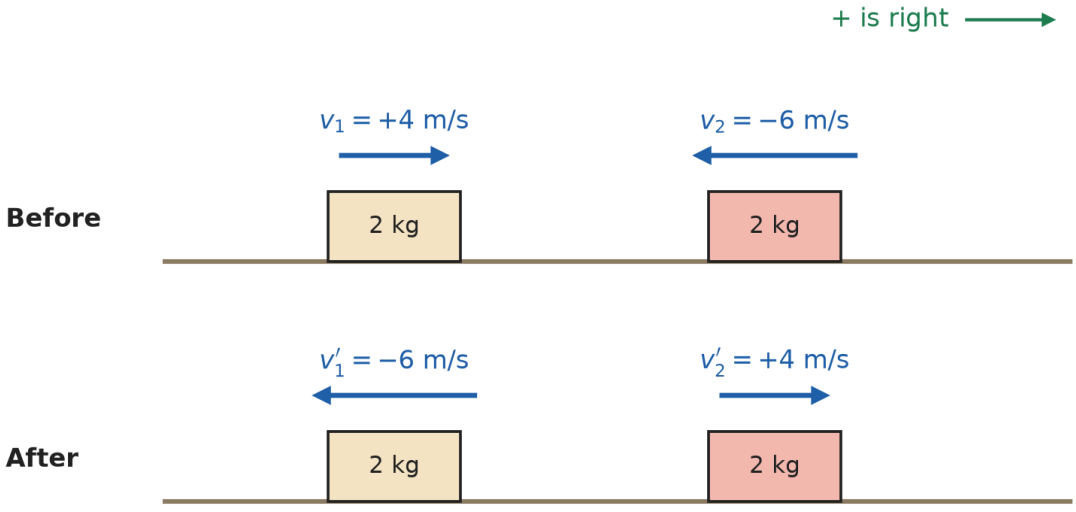
2. Two objects of equal mass: Two 2 kg carts are approaching each other, one at 4 m/s [right]and the other at 6 m/s [left]. Determine the velocity of each cart after the elastic collision.

GIVEN

$m_1 = 2$ kg	$v_1 = +4$ m/s	$m_2 = 2$ kg
$v_2 = -6$ m/s	$v_1' = ?$	$v_2' = ?$

Sign convention: right is +.

Initial and final state (+ is right)



Step 1 – The 2 kg cart on the left

Use the lesson's equations for a head-on elastic collision.

Formula: $v_1' = \left(\frac{m_1 - m_2}{m_1 + m_2}\right)v_1 + \left(\frac{2m_2}{m_1 + m_2}\right)v_2$

Put in the numbers: $v_1' = \left(\frac{2-2}{2+2}\right)(4) + \left(\frac{2(2)}{2+2}\right)(-6)$

$v_1' = (0.0000)(4) + (1.0000)(-6) = 0.000 - 6.000 = -6.000 \text{ m/s}$

Step 2 – The 2 kg cart on the right

Formula: $v_2' = \left(\frac{m_2 - m_1}{m_1 + m_2}\right)v_2 + \left(\frac{2m_1}{m_1 + m_2}\right)v_1$

Put in the numbers: $v_2' = \left(\frac{2-2}{2+2}\right)(-6) + \left(\frac{2(2)}{2+2}\right)(4)$

$v_2' = (0.0000)(-6) + (1.0000)(4) = 0.000 + 4.000 = 4.000 \text{ m/s}$

FINAL ANSWER

$v_1' = -6.0$ m/s; $\vec{v}_1' = 6.0$ m/s [left]

$v_2' = +4.0$ m/s; $\vec{v}_2' = 4.0$ m/s [right]

Check: equal masses trade velocities (special case B): the cart that had +4 m/s now has −6 m/s, and the other has +4 m/s.

PRACTICE QUESTION 3

One object initially at rest

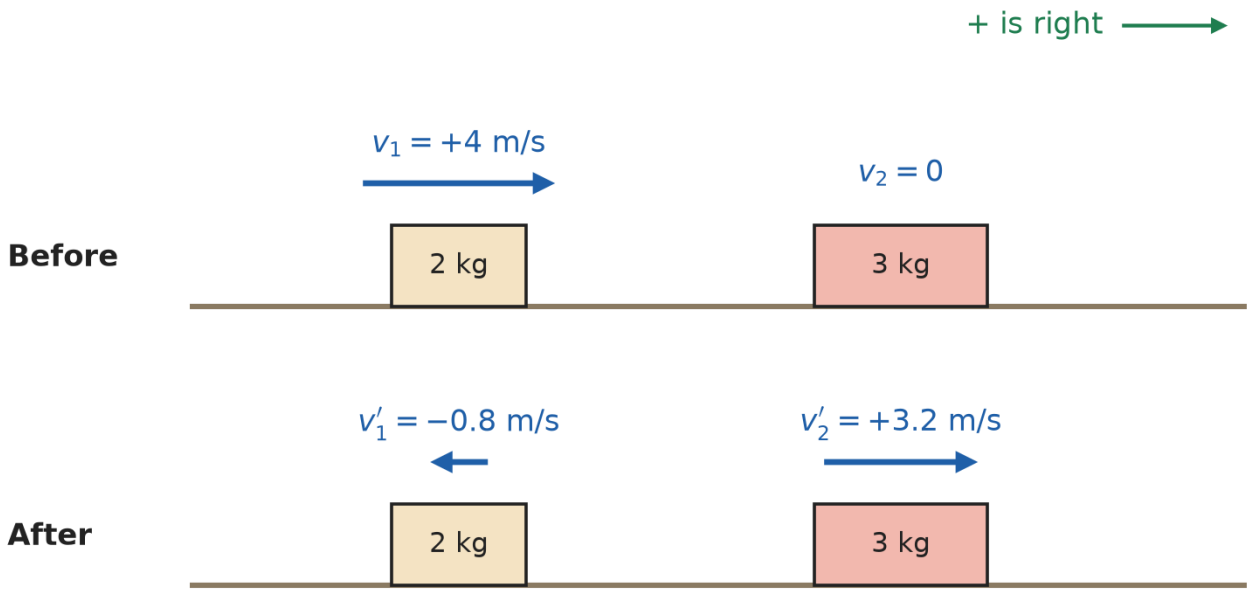
3. One object initially at rest. A 2 kg ball rolls towards a stationary 3 kg ball at 4 m/s [right]. Determine the velocity with which each ball will roll away after the elastic collision.

GIVEN

$m_1 = 2$ kg	$v_1 = +4$ m/s	$m_2 = 3$ kg
$v_2 = 0$ m/s	$v_1' = ?$	$v_2' = ?$

Sign convention: right is +.

Initial and final state (+ is right)



Step 1 – The 2 kg ball on the left

Use the lesson's equations for a head-on elastic collision.

Formula: $$v_1' = \left(\frac{m_1 - m_2}{m_1 + m_2}\right)v_1 + \left(\frac{2m_2}{m_1 + m_2}\right)v_2$$

Put in the numbers: $$v_1' = \left(\frac{2-3}{2+3}\right)(4) + \left(\frac{2(3)}{2+3}\right)(0)$$

$$v_1' = (-0.2000)(4) + (1.2000)(0) = -0.800 + 0.000 = -0.800 \text{ m/s}$$

Step 2 – The 3 kg ball on the right

Formula: $v_2' = \left(\frac{m_2 - m_1}{m_1 + m_2}\right)v_2 + \left(\frac{2m_1}{m_1 + m_2}\right)v_1$

Put in the numbers: $v_2' = \left(\frac{3-2}{2+3}\right)(0) + \left(\frac{2(2)}{2+3}\right)(4)$

$v_2' = (0.2000)(0) + (0.8000)(4) = 0.000 + 3.200 = 3.200$ m/s

FINAL ANSWER

v_1' = −0.80 m/s; $\vec{v}_1'$ = 0.80 m/s [left]

v_2' = +3.2 m/s; $\vec{v}_2'$ = 3.2 m/s [right]

Check: momentum before = (2)(4) = 8 kg·m/s; after = (2)(−0.8) + (3)(3.2) = −1.6 + 9.6 = 8 kg·m/s.

PRACTICE QUESTION 4

Catching up

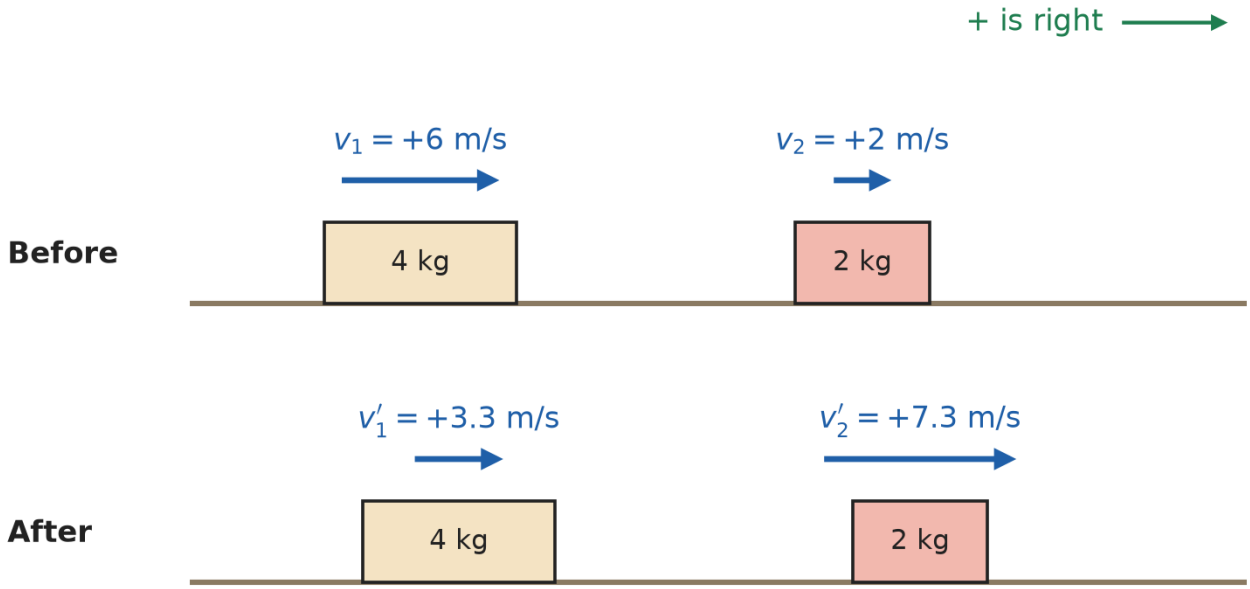
4. A 4 kg ball rolls at 6 m/s to the right. A 2 kg ball is ahead of it, rolling at 2 m/s, also towards the right. The 4 kg ball eventually catches up to and collides elastically with the 2 kg ball. What will the velocities of the two balls after the collision?

GIVEN

$m_1 = 4$ kg	$v_1 = +6$ m/s	$m_2 = 2$ kg
$v_2 = +2$ m/s	$v_1' = ?$	$v_2' = ?$

Sign convention: right is +.

Initial and final state (+ is right)



Step 1 – The 4 kg ball on the left

Use the lesson's equations for a head-on elastic collision.

Formula: $$v_1' = \left(\frac{m_1 - m_2}{m_1 + m_2}\right)v_1 + \left(\frac{2m_2}{m_1 + m_2}\right)v_2$$

Put in the numbers: $$v_1' = \left(\frac{4-2}{4+2}\right)(6) + \left(\frac{2(2)}{4+2}\right)(2)$$

$$v_1' = (0.3333)(6) + (0.6667)(2) = 2.000 + 1.333 = 3.333 \text{ m/s}$$

Step 2 – The 2 kg ball on the right

Formula: $v_2' = \left(\frac{m_2 - m_1}{m_1 + m_2}\right)v_2 + \left(\frac{2m_1}{m_1 + m_2}\right)v_1$

Put in the numbers: $v_2' = \left(\frac{2-4}{4+2}\right)(2) + \left(\frac{2(4)}{4+2}\right)(6)$

$v_2' = (-0.3333)(2) + (1.3333)(6) = -0.667 + 8.000 = 7.333 \text{ m/s}$

FINAL ANSWER

v_1' = +3.3 m/s; $\vec{v}_1'$ = 3.3 m/s [right]

v_2' = +7.3 m/s; $\vec{v}_2'$ = 7.3 m/s [right]

Check: momentum before = (4)(6) + (2)(2) = 28 kg·m/s; after = (4)(3.333) + (2)(7.333) = 28 kg·m/s.